Legionnaires' × Cooling-Tower History — Findings

2026 Upper East Side (UES) Legionnaires' disease community cluster Analysis date: 2026-07-17 · Data: NYC Open Data (Socrata) + DOH press-release list

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1. Question and framing

Is there a relationship between the NYC buildings implicated in the 2026 UES Legionnaires' cluster and those buildings' cooling-tower **registration, Legionella-sampling, and inspection history**?

The DOH "clean & disinfect" list is a roster of **buildings whose cooling tower(s) tested PCR-positive for *Legionella*** during the July 2026 investigation — not patient addresses (those are never published). So this is a **case–control study of cooling towers**:

> **Outcome:** building's tower(s) PCR-positive & ordered to remediate (case=1) vs not (control=0)
> **Exposures:** historical compliance / maintenance features engineered from the two open datasets.

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2. Data

Dataset	Socrata ID	Rows pulled
NYC Cooling Tower Registrations (+ Legionella sample dates)	`y4fw-iqfr`	5,949
NYC Cooling Tower System Inspection Results	`f9wb-g8mb`	122,846
DOH affected-building list (31 on 07-10 + 45 update)	press release	**76 buildings**

All 76 case addresses matched to a registered building (BIN) after address normalization — **0 unmatched**. (Sanity checks: 1000 Fifth Ave = 21 towers = the Met; "300 E 83rd", flagged "unregistered" by DOH, does appear in registrations.)

Leakage control: every history feature is capped at **2026-07-01** — the July-2026 investigation samples *are* the outcome, so including them would leak it.

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3. Design & sample

Positivity is overwhelmingly a function of **being in the sampled area**:

Area	Buildings	Positivity
Outbreak ZIPs (10028 / 10075 / 10128)	183	**40.4%**
Rest of Manhattan	2,699	0.07%

Buildings outside the zone were never sampled, so counting them as "controls" would be surveillance bias. **Primary analysis is therefore restricted to the 183 registered buildings in the outbreak ZIPs (74 cases, 109 controls).**

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4. Results

4a. Univariate (case vs control, in zone)

Significant / notable (Mann-Whitney or Fisher):

Feature	Case (median)	Control (median)	p
Registration age (yrs)	10.8	10.8	**0.003**
# Legionella samples (24 mo)	4	3	**0.010**
Median gap between samples (days)	41.5	54.5	**0.025**
Days since last sample	16	20	0.27
# Violations (all-time)	17	16	0.37
Ever cited for a violation	93%	94%	1.00
Overdue >90d on sampling	9%	18%	0.14

Direction is the opposite of a naïve "negligence" hypothesis: case towers were sampled **more** often and had **shorter** gaps, not fewer. Violation/citation history did **not** differ.

4b. Multivariable logistic regression (primary model)

183 buildings, 74 cases, 6 predictors, all VIF < 2.2. ORs are per **1 SD** (continuous). Figure: `reports/figures/odds_ratios.png`.

Predictor	OR	95% CI	p
Registration age	**1.57**	0.99–2.49	0.053
# samples (24 mo)	**1.49**	0.94–2.35	0.089
# violations	1.13	0.78–1.63	0.51
Days since last inspection	0.81	0.58–1.12	0.20
# towers	1.00	0.65–1.54	0.99
Days since last sample	0.96	0.63–1.46	0.86

Model: McFadden $R^2 = 0.065$, in-sample AUC = 0.70, joint LLR $p = 0.013$.

L1/LASSO companion (guards against overfitting): keeps the same two leading signals — registration age (OR ≈ 1.53) and sampling frequency (OR ≈ 1.46) — shrinks # towers to zero. Cross-validated AUC ≈ 0.60 (weak).

4c. Spatial autocorrelation

Moran's I (binary weights, 400 m): case indicator **I = 0.0004, p = 0.15**; model residuals **I ≈ 0, p = 0.38**. **No fine-scale spatial clustering** of positive towers within the zone — positives and negatives are intermixed (`reports/figures/map_towers.png`).

4d. PLUTO sensitivity (building attributes)

Joining PLUTO by BBL (173/183 zone buildings matched), **case buildings are physically bigger and higher-valued**, not older:

Attribute	Case (median)	Control (median)	p
Assessed total value	\$21.2M	\$14.7M	**0.004**
Building floor area	144,280	106,034	**0.025**
# Floors	18	16	**0.046**
Building age (yrs)	62	63	0.48

Re-running the logistic model with PLUTO covariates added (`reports/odds_ratios_pluto.csv`): the **registration-age signal attenuates to null** (OR 1.20, p=0.38) — it was partly a proxy for building size/value — while **sampling frequency stays significant** (OR 1.52, p=0.023) and assessed value trends positive (OR 1.49, p=0.091). Model AUC 0.71, R^2 0.072. So the most robust correlate of positivity is *how much a tower is sampled*, plausibly detection intensity, with building size/value as a secondary axis.

4e. Tier-1 additional predictors

Nine further hypotheses were tested (`src/tier1.py`, `reports/odds_ratios_tier1.csv`): severity-split violations (PHH / Critical), Legionella / water-quality–specific violations, follow-up ("non-cycle") inspections, local cooling-tower density within 200 m, sampling-cadence regularity, and ever-inactive status.

New predictor	Case vs control	p
Ever had a **follow-up ("non-cycle") inspection**	35% vs 20%	**0.027**
# non-cycle inspections	higher in cases	**0.015**
Water-quality / Legionella violations (focused model)	OR 1.68 / SD	0.106
Local tower density (200 m)	10 vs 10	0.60 (null)
PHH / Critical violation counts	—	0.59 / 0.24 (null)
Sampling-cadence irregularity (gap CV)	0.82 vs 0.68	0.15 (null)
Ever-inactive tower	4% vs 6%	0.74 (null)

The one nominal signal is **follow-up inspections** — complaint- or problem-triggered visits, i.e. buildings previously flagged. In the focused 6-predictor model (n=183) **sampling frequency**

remains the most robust correlate (OR 1.58, p=0.017); water-quality violations trend positive but are not significant; AUC 0.66.

> **Multiple-comparison caveat:** ~9 features were tested, so ~0.5 false positives are expected at α=0.05. The non-cycle signal (p≈0.02) does **not** survive a Bonferroni correction (~0.006) and should be read as a lead to investigate, not a finding. Tower density being null is itself informative: positive towers are **not** concentrated where towers are densest.

4f. Tier-2 additional data (LL84 energy/water · DOB · HPD)

Pulled three more open datasets and joined by BBL to the zone (`src/tier2.py`): LL84 benchmarking (`5zyy-y8am`; energy/water use as a cooling-load proxy), DOB violations (`3h2n-5cm9`), and HPD housing violations (`wvxf-dwi5`).

New predictor	Case vs control (median)	p
# DOB violations (all-time)	53 vs 35	**0.0004**
Site energy-use intensity (EUI)	82.5 vs 72.6	0.094
Water use (kgal) — *tower makeup-water proxy*	4,670 vs 4,214	0.33 (null)
Water-use intensity (kgal/ft²)	—	0.85 (null)
Energy Star score	44 vs 40	0.62 (null)
# HPD violations / Class C (hazardous)	3 vs 3 / 0 vs 0	0.68 / 0.41 (null)

DOB violations are the strongest raw signal in the whole study — and a textbook confounder. They vanish once building size is controlled:

Model	DOB odds ratio / SD	p
DOB alone	1.60 (1.06–2.43)	0.026
DOB + assessed value + # towers	1.22 (0.74–2.01)	0.43
+ building age	1.18 (0.78–1.79)	0.44

Bigger, older buildings simply accumulate more DOB violations — so the "positive buildings have more violations" pattern is a **size artifact, not negligence**, the same confound seen throughout (§4d). Notably, the **cooling-load proxies themselves — water use and energy intensity — did not hold up** (water null; EUI attenuates after adjustment), so heavier air-conditioning use is not a clear driver either. The focused Tier-2 model reaches AUC 0.75 but on a reduced sample (n=103, LL84 coverage), and the durable signals are still detection-related (sampling frequency; recency of inspection).

4g. Tier-3: construction proximity (null)

Current DOB NOW permits (`rbx6-tga4`, `src/tier3_construction.py`) in the UES bounding box: **45,020 permits**. The neighborhood is so densely permitted that every building sits a median **~740**

permits within 150 m. Distance to nearest construction (p=0.57) and permit density within 150 m (p=0.28) were both null — in a saturated area, construction proximity can't discriminate cases from controls.

4h. Full predictor inventory (~40 variables tested)

Domain	Variables tested	Verdict
Tower testing history	# samples (12/24 mo), days since last sample, sampling gaps + irregularity, overdue-90d, never-sampled	null — except **sampling *frequency*** (↑ in cases, detection signal)
Tower inspections	# inspections, days since last, cycle vs **non-cycle**, ever-inactive	null — except **follow-up (non-cycle)** inspections (weak, not MC-robust)
Tower violations	total, ever-cited, # types, **Critical**, **PHH**, **Legionella/water-quality**	**all null**
Building physical (PLUTO)	year built/age, # floors, res/total units, **floor area**, **assessed value**	size/value ↑ in cases → **confounder**, not negligence
Energy / water (LL84)	site & source EUI, **water use**, water intensity, Energy Star	null / weak (cooling-load proxy did **not** hold)
Management quality	**DOB violations**, HPD violations, HPD Class-C	DOB significant alone but **erased by size adjustment**; HPD null
Spatial / context	outbreak-ZIP, **tower density (200 m)**, Moran's I, **construction proximity**	ZIP dominates; within-zone density & clustering **null**

Two things survive everything: *location* (being in the sampled zone) and *detection intensity* (how often a tower is looked at). Every negligence-flavored variable is null or dissolves under size adjustment.

4i. Tier-4 advanced designs & angles

Beyond more predictors, we ran different *questions and study designs* (`src/tier4_*.py`):

Angle	Method	Result
#7 Matched case-control	63 case↔control pairs matched on size/value/age; paired tests on compliance	**Confirms negligence-null**: violations still null ($p=0.85$); only **more sampling** ($p=0.026$) and **more recent inspection** ($p=0.009$) survive — detection signals, not maintenance
#3 Directional gradient	trend-surface logistic on projected coords	**New (weak) signal**: positivity rises to the **north** (N–S OR 2.0/km, $p=0.043$; bearing ~336°; joint $p=0.068$) — a gradient Moran's I is blind to
#8 "Lottery" model	binomial P(pos)=1−(1−p)^towers vs constant	Lottery **rejected** (LR $p=0.44$); tower *count* doesn't drive positivity → per-tower conditions do
#1/#2 Owner/contractor network	shared PLUTO owner + DOB permit contractor across the 76	**Null** — buildings are single-asset LLCs; shared contractors (scaffolding/mechanical) appear equally on controls. *Water-treatment vendor — the relevant one — isn't in open data*
#5 Repeat-offenders / prior clusters	prior Legionella violations; past NYC cluster geographies	**Null / no overlap** — Legionella-specific violations flat (§4e); prior NYC clusters were in the Bronx/Upper Manhattan (different geography)
#6 Weather trigger	July-2026 timeline vs conditions	**Consistent** with an environmental bloom — the cluster coincided with a heat wave that officials linked to Legionella growth (qualitative)
#9 Real outcome / tower hardware	NYS registry `gd58-9fej`; NYC results fields	**Unavailable** — datasets are dates-only (no results); NYS registry is a 74-row upstate sample with no NYC coverage, so manufacturer/model/capacity can't be joined
#4 Aerosol plume model	Gaussian plume vs case locations	**Not feasible** on open data — requires patient coordinates (private); the #3 gradient is the available substitute

Takeaways: (1) the negligence-null now survives a *matched* design — the cleanest test — so it is not a size artifact; (2) the only genuinely new positive result is a **weak northward gradient** in positivity, worth noting but borderline and unadjusted for multiple looks; (3) every mechanistic link we *could* build from open data (ownership, contractors, tower hardware, prior history) is null or unavailable, reinforcing that the deciding variables are tower-level and unpublished (§5b).

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5. Interpretation

1. **The dominant "predictor" of a positive tower is geography, not maintenance history** — 40% positivity in-zone vs ~0% elsewhere. This is consistent with a diffuse neighborhood aerosol exposure, reinforced by the flat Moran's I (no single point-source hotspot within the zone).

2. **Within the outbreak zone, compliance history explains little** (AUC 0.70, R^2 0.07). Crucially, the metrics a regulator would flag — violations, citations, overdue sampling, sparse inspections — were **not** associated with testing positive. This argues the cluster is **not** simply a "negligent operators" story.
3. The two weak positive associations both plausibly reflect **detection/confounding, not causation**: towers sampled more frequently are more likely to *catch* a positive (surveillance intensity), and older registrations proxy for older/larger building systems. Neither is evidence that more testing *causes* Legionella.

Bottom line: there is a modest, statistically borderline relationship between a building's cooling-tower history and outbreak positivity, but it is weak, points opposite to a negligence hypothesis, and is dwarfed by location. The data do not support singling out poorly-maintained towers as the driver of this cluster.

5b. Why positivity isn't "random" — the variables open data can't see

"Not explained by our variables" is **not** the same as "random." Three real, non-random drivers *are* in the data: **location**, **building scale**, and **detection intensity**. What's genuinely flat is the *negligence/compliance* axis.

The reason the within-zone pattern still looks like scatter is that **whether a given tower blooms is decided by tower-level operational conditions that no public dataset records.** The registration data lists the *dates* a tower was sampled — never the *results*. The deciding factors live one layer below open data:

- **Water temperature** in the basin (the dominant growth factor)
- **Biocide/treatment dosing** and any lapse in it
- **Stagnation** — short idle periods, dead legs, a pump cycling off
- **Biofilm / scale**, and **drift-eliminator** condition (how much mist escapes)
- **Nutrient load** and makeup-water quality
- **Exact timing** of the last real cleaning relative to the July heat
- **Tower make / model / design** — some aerosolize far more than others

Two buildings identical on every public field can differ entirely on these — and *that* is what separates a positive tower from a negative one. Legionella clusters are characteristically idiosyncratic per-tower plus weather plus chance; this kind of within-area scatter is exactly what the epidemiology predicts.

Power: with 74 cases we can only detect *large* effects; a real OR≈1.3 would need several hundred cases. "Null" means "no large effect visible in building-level open data," not "no effect exists." We are also using **building-level administrative records** to chase a **tower-level microbiological** event — an inherent ceiling.

What would actually explain which towers bloomed (and what DOH uses internally, unpublished): per-tower *Legionella* culture counts over time; water-treatment and maintenance logs; tower make/model/age; and — the gold standard — **genomic matching** of each tower's isolate to patient isolates. A regression on open data structurally cannot name the source; it can only rule hypotheses in or out, which is what we did.

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6. Limitations

- Outcome = **tower positivity**, not human infection (patient addresses private).
- Small case count (74 in-zone) → wide CIs; borderline p-values are fragile.
- **Surveillance/detection bias:** positives were actively *sought* in these ZIPs.
- ~6-month posting lag on inspections; owner-reported sample dates may be imperfect.
- Registration age is a proxy (date first registered ≈ 2015 floor), not true tower age.
- Association ≠ causation; ecological/aggregation caveats apply.

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7. Reproducibility

```
python3 src/fetch_data.py        # pull Socrata datasets -> data/raw/
python3 src/build_case_list.py   # 76-building case roster -> data/raw/
python3 src/match_and_label.py   # address match/audit -> data/processed/
python3 src/build_features.py    # labeled feature table -> data/processed/
python3 src/analyze.py           # EDA + regression + spatial -> reports/
python3 src/enrich_pluto.py      # PLUTO join + sensitivity model -> reports/
python3 reports/build_map.py     # self-contained artifact map (NYC grid basemap)
python3 reports/build_leaflet.py # standalone map w/ OpenStreetMap tiles (open locally)
```

Console logs: `reports/stats.txt`, `reports/stats_pluto.txt`. Odds ratios: `reports/odds_ratios.csv`, `reports/odds_ratios_pluto.csv`.

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8. Interactive maps

- **`reports/map.html`** — self-contained interactive dot-map over the authentic NYC street grid (basemap from NYC Open Data `inkn-q76z`); hover history, filters, table view; works inside a sandboxed artifact (no external requests).
- **`reports/map_leaflet.html`** — standalone map on a live OpenStreetMap tile basemap; open in a browser locally (loads external tiles, so not for the sandbox).

When the DOH list grows further, drop the new addresses into `src/build_case_list.py` and re-run — the pipeline is a drop-in re-run.